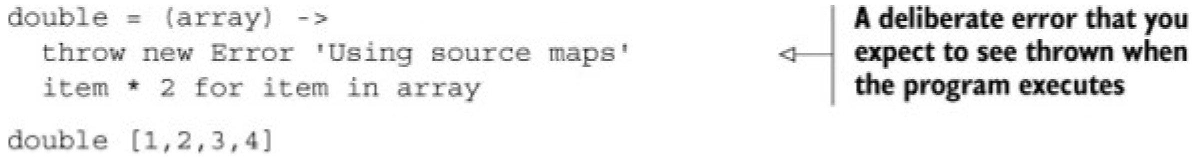
To understand what they do, here's a trivial program you can whet your source-maps appetite on:



```
double = (array) ->
  throw new Error 'Using source maps'
  item * 2 for item in array

double [1,2,3,4]
```

A deliberate error that you expect to see thrown when the program executes

Suppose the program is contained in a file called sourcemaps.coffee. You're familiar with the standard way to compile:

```
> coffee sourcemaps.coffee
```

To generate a corresponding *map* file, you pass the `-m` or `--map` flag:

```
> coffee --map sourcemaps.coffee
```

The source map won't make much sense because it's not intended to be human readable. It's a JSON file containing information that the runtime can use to map from the compiled JavaScript program to your original CoffeeScript source:

```
{
  "version":3,
  "file":" sourcemaps.coffee",
  "sources":[
    "sourcemaps.coffee"
  ],
  "names":[],

"mappings":"AAAC;;;EAAA,MAAA,GAAS,SAAA,CAAA,KAAA,CAAA;;;MAAW,2BAAqB
,aAArB,
aAAA,CAAA,KAAA,CAAA;QAAa,OAAQ;oBAArB,IAAA,CAAA,CAAA,CAAO;;;;;EAC3B,
OAAO,IAAP,CAAY,MAAA,CAAO,CAAA;AAAA,IAAC,CAAD;AAAA,IAAG,CAAH;AAAA,IA
AK,CAAL;
AAAA,IAAO,CAAP;AAAA,EAAA,CAAP,CAAZ;EACA,KAAA,CAAM,GAAA,CAAI,KAAJ,CA
AU,
mBAAV,CAAN"
}
```

To use the source map, the runtime needs to be told where to find it. To tell the runtime where to find the source map, either put a comment in the